

Course Title	Discrete Mathematics		
Course Code	SC205	Credit Structure	3-1-0-4 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 25-26)
Program	B.Tech		
Prerequisites	--		
Course Objectives/ Brief Course Description	Discrete Mathematics covers the concept of mathematical proofs, concrete proof techniques, and the use of mathematical language to make precise statements and draw accurate and insightful conclusions on discrete mathematical structures of a wide variety.		
Evaluation/ Grading Policy	Tutorials: 20%, Insem 1: 20%, Insem 2: 20%, Final: 40%		
Course Materials/ References	Discrete mathematics and its applications, Kenneth H Rosen, Tata McGrawHill, 2011. Elements of Discrete Mathematics C. L. Liu, Tata McGrawHill		
Detailed Course Content	Propositional Logic, First Order Logic, Sets, Relations, Functions, Proof Techniques, Mathematical and Structural Induction, Recursion and Recurrence relations, Counting Techniques, Binomial Theorem, Permutations & Combinations, Matrices, Number Theory, Computational Problems, Algorithms, Formal Languages, Computational Models, Graphs, Trees.		

Course Outcome: After completion of the course, the students should acquire:

- A basic overview of discrete mathematics [PO1, PO12]
- Understand how the concepts are used in various applications [PO3, PO4, PO5].
- Apply skills learned in the course for problem solving [PO2, PO3, PO5]

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X							X

Course Outcomes and Competencies

Sr. No.	Course Outcome	PO/PSO	CPD	KD
1.	CO1: A basic overview of discrete mathematics	PO1,PO12/PSO1	U	F, C
2.	CO2:Understand how the concepts are used in various applications	PO3, PO4, PO5/ PSO1, PSO2	U	F, C, P
3.	CO3: Apply skills learned in the course for problem solving	PO2, PO3, PO5/ PSO1, PSO2	U, Ap	F, C, P

CL: Cognitive Process Domain [R: Remember; U: Understand; Ap: Apply; An: Analyze; E: Evaluate; C: Create]

KC: Knowledge Domain [F: Factual; C: Conceptual; P: Procedural; M: Metacognitive]

Focus Area	Yes/No	Details
Focus on Employability	Yes	The students taking the course can be employed in industries focusing on software development.
Focus on Skill development	Yes	The students develop necessary skills to develop foundational knowledge in discrete mathematics that become prerequisite for other subsequent courses.
Focus on entrepreneurship	No	-